Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Semester II, 2025-2026

Tutorial-10 (Introduction to the Digital World, Logical expressions, and Hardware implementation)

Before a processor can compute, it needs a way to represent numbers and deal with physical hardware limitations (like finite bit-widths and round-off errors).

1. Find the decimal equivalent of the following binary numbers.

   (a) 1001101 (b) 11100011 (c) 10101010

2. Find the decimal equivalent of the following HEX numbers

   (a) F4F (b) FED (c) FEED

3. Find the binary representation of decimal fraction 0.65 to 8 bit. Is there a round of error? How large is it?

An ALU doesn't usually have a dedicated subtractor. It uses 2's complement addition to do both. Please check if there is an overflow.

4. Convert the following 2’s complement numbers to their signed decimal equivalents. (a) 11111111 (b) 00010000 (c) 10000001

5. Show how the following can be added in 2’s complement notation using 8-bit arithmetic. (a) (-1) +45 (b) 128+(-60) (c) 55+75

How do we take a practical requirement and turn it into a truth table and corresponding logic expression?

6. Devise a logic circuit to sound a buzzer B whenever the car ignition key is inserted and either the seat belt of the driver (D) or the passenger (P) is unfastened. Write down the truth table (TT) and the logic function. If the problem is modified to include a pressure sensor to provide information about seat occupancy, find the resulting TT and logic function.

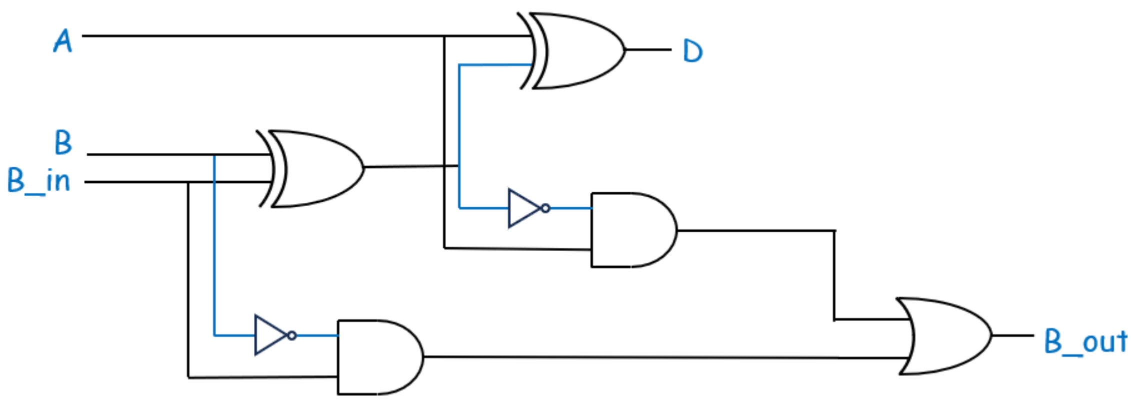
ALUs must be fast and physically small. We use k-maps to reduce the gate count and delay.

7. Use k-maps to obtain MSOP and MPOS for each of the following functions:

(a) $Z = \overline{A}\,\overline{B}\,\overline{C}D + \overline{A}B\overline{C}D + AB\overline{C}D + A\overline{B}\,\overline{C}D + A\overline{B}\,\overline{CD}$ with don’t care for $ABCD = 1010$

(b) $Z = \left(\overline{A} + B + \overline{C}\right)\left(A + B + \overline{C}\right)$ with don’t cares for ABC = 111 and 110

Processors execute math column by column. Whether adding or subtracting, a multi-bit ALU requires logic circuits that can process the current bits while accounting for information carried over from the previous column.



8. For the circuit shown above, fill in all the intermediate signal names and find simplified logic expressions for D and B_out. Construct a Truth Table (TT) for this circuit using inputs A, B, and B_in. Based on the TT, deduce what exactly do D and B_out represent?

Real ALUs are frequently built using universal gates to maximize manufacturing efficiency. The following problem bridges theory and physical hardware by having you map an optimized equation directly onto standard, off-the-shelf integrated circuits.

9. A combinational circuit has four inputs (A, B, C, and D) and an output Z. The output is 1 whenever three or more of the inputs are zero, otherwise the output is zero.
    a. Find the minimum sum-of-products (MSOP) expression for Z.
    b. Design the combinational circuit for this expression using only 7404 inverters, 7400 2-input NAND gates, and 7410 3-input NAND gates.

10. In Q8, you analyzed a dedicated logic circuit for subtraction. However, real ALUs save valuable hardware space by reusing the exact same circuit for both addition and subtraction. It is achieved by applying the 2's complement math you practiced in Q5.

a. Construct the Truth Table and derive the simplified Boolean expressions for the Sum (S) and Carry-out (C_out) of a standard 1-bit Full Adder (Inputs: A, B, C_in).
b. Assume you have a mode control signal, M. When M=0, the ALU performs Addition (A+B). When M=1, it performs Subtraction (A–B by performing A+B+1). Using only your Full Adder from part (a) and an additional logic gate, explain how you would wire M, B, and the initial C_in to toggle between these two operations without ever needing the dedicated subtractor from Q8.